

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 - Digital Electronics
Spring Term 2026

Lab. 12-SRAM Read

Issued: 4/22/2026

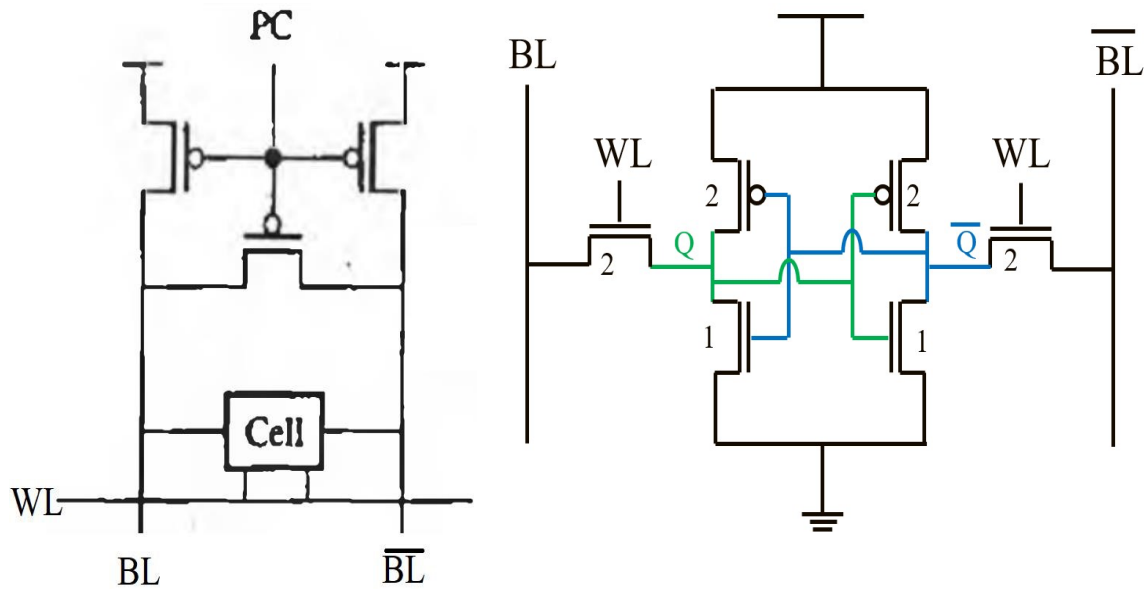
Due: 4/24/2026

Question 1:

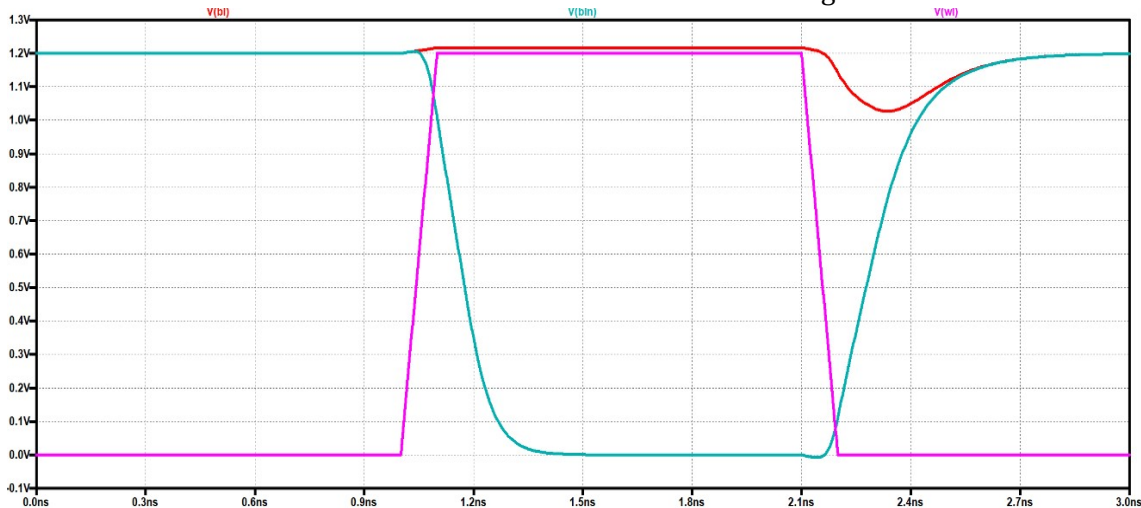
Objectives:

- SRAM Read

Given the SRAM cell on the right, and Pre-charge circuit for the SRAM on the left:

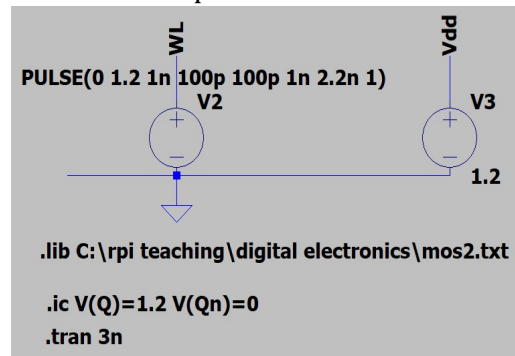


- i. First, we will check this **Q and Q'** values by hand calculations:
 - i. First, determine which transistors are on in this scenario.
 - ii. Next, determine their operating region.
 - iii. Last, calculate **Q and Q'**.
 1. Perform calculations assuming the worst-case scenario (WL transistors are fully ON, while the **BL'** voltage has not yet changed).
- ii. Next, Build the circuit shown in LTSPICE.
 - i. Circuit will use 130nm process parameters as defined in **MOS2.txt**.
 1. $L = 100\text{nm}$ for all transistors.
 2. W/L ratio is given on SRAM schematic.
 - ii. Include 10fF capacitors to GND on **BL** and **BL'**.
- iii. Perform a 3ns transient simulation.
 - i. Set an initial value of logic 1 in the SRAM.
 - ii. Pre-charge bit-lines to logic 1 (1.2V) for 1ns.
 - iii. Turn off pre-charge, then read the SRAM for an additional 1ns.
 - iv. Disable read and reassert pre-charge for a final 1ns.
 - v. The transient simulation will look like the following:



- vi. You only require 2 voltage sources.
 1. $V_{DD} = 1.2\text{V}$
 2. WL
 - a) Since we are doing read with just one SRAM cell, the precharge signal, **PC**, can be shared with **WL**. The Pre-charge is PMOS, while the Word-line is NMOS, and thus they will naturally be inverted.
 - b) Set **WL** to be a **VPULSE** source. Have it initially be 0 for 1ns, then 1.2V from 1ns $\rightarrow$ 2ns, then fall back to 0 for 2ns $\rightarrow$ 3ns. Use 100ps rise and fall times.
- vii. Initial conditions will need to be set on **Q** and **Q'** to set an initial stored value. Set the logic cell to be logic 1.

viii. Your LTSPICE setup should look like the following:



ix. Perform the transient simulation. Record the following for the duration:

1. BL
2. BL'
3. WL
4. Q
5. Q'

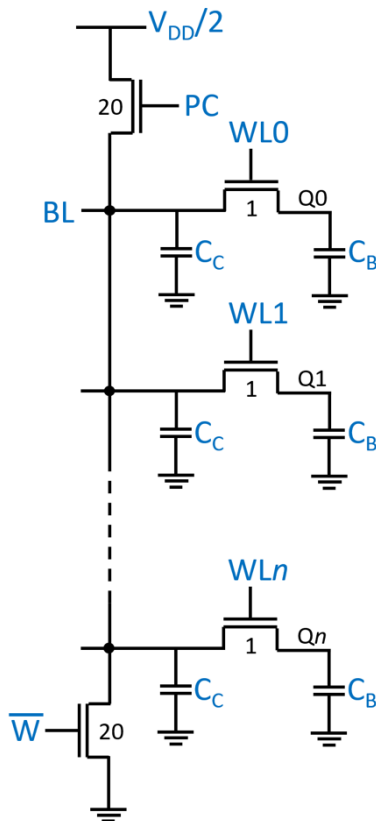
iv. While reading, state the maximum voltage **Q'** rises to.

Question 2:

Objectives:

- DRAM Design

You are to implement a column in DRAM with as many cells possible with the topology on the left, and the parameters on the right. Transistor W/L ratios are present in the schematic.



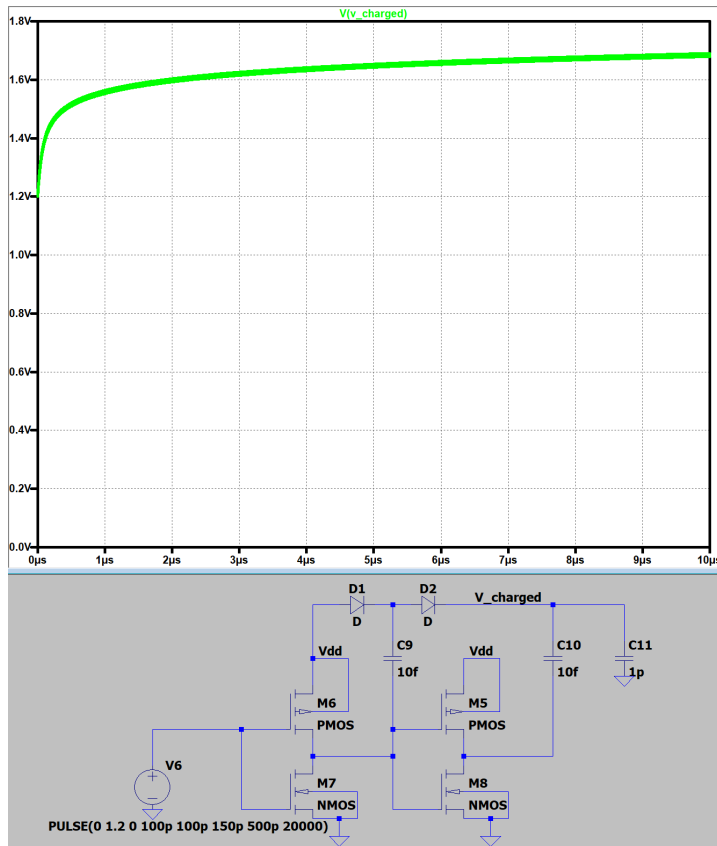
- $V_{DD} = 1.2 \text{ V}$
- $C_B = 25 \text{ fF}$
- $C_C = 1 \text{ fF}$
- $V_{WL}(\text{logic } 1) = 1.7 \text{ V}$
- $V_{BL}(\text{logic } 1) > 700 \text{ mV}$
- $V_{BL}(\text{logic } 0) < 500 \text{ mV}$
- $L = 100 \text{ nm}$

- Note that the Pre-charge is now handled by an NMOS transistor. This is allowable since pre-charge in DRAM only pulls up to $V_{DD}/2$.
- For this DRAM, charge boosting is used to elevate the word line signals to 1.7V, and thus allowing the full $V_{DD} = 1.2 \text{ V}$ to be stored in the DRAM capacitor, C_B .
 - This results in a simplification in calculation of ΔV_{BL} on read, which is:
 - $$\Delta V_{BL} = \frac{C_B}{C_B + \Sigma C_C} * \frac{V_{DD}}{2}$$

- To support the sense amplifier, V_{BL} must be less than $V_{DD}/2 - 100 \text{ mV}$ to read logic 0, and V_{BL} must be greater than $V_{DD}/2 + 100 \text{ mV}$ to read logic 1. Calculate the total number of DRAM bits that can be present in one column with $C_B = 25 \text{ fF}$, and $C_C = 1 \text{ fF}$.
- Demonstrate both read 0 and read 1 in LTspice by setting a load capacitor on the bit-line that is equal to ΣC_C when the ΔV_{BL} requirement is barely met. Only one DRAM cell needs to be simulated. For read of logic 0 and logic 1, two separate simulations can be performed with different initial values, or by reading 1, writing 0, then reading 0.
 - Comment on how the simulation compares to your hand calculated results.

Charge Boost Amplifier

This circuit does not need to be built for this lab, but this creates the 1.7V voltage required by word-line. In lab you can just specify word-line as having a logic high of 1.7V.



- $V_{\text{clock}} = 2 \text{ GHz}$
- $V_{\text{charged}} = 2 \cdot V_{\text{DD}} - V_{\text{Diode}}$

Report Submission:

Submit your work on LMS under **Labs**

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTspice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters.**
- Due date: Friday at 11:59pm.